

CROSS-BASE CORRELATIONS OF ARTIN PRIMES: ENTANGLEMENT, EXCLUSION, AND THE STRUCTURE OF $p - 1$

ABSTRACT. Say a prime p is *Artin base* a if a is a primitive root modulo p . This paper studies the joint behaviour of Artin status *across bases* at the same prime. We first record a deterministic exclusion law: for *any* two bases a, b , writing $d = \text{sqf}(ab)$ for the squarefree part of ab , no prime p with $\left(\frac{d}{p}\right) = -1$ is simultaneously Artin for a and for b . That condition is a union of residue classes modulo $4d$, so whenever $d > 1$ — equivalently, whenever a and b do not have the same squarefree part — half of all primes are barred outright; the proof is two lines from the multiplicativity of quadratic characters. A corollary is the triple form of the law: if $\text{sqf}(c) = \text{sqf}(ab)$ then no odd prime is simultaneously Artin for a , b and c — for instance, no odd prime has 2, 5 and 10 all as primitive roots. Over all 66 pairs of the base set below and all 50,847,524 primes below 10^9 , the number of primes violating the pair law is zero, as it must be. We then measure, for all 66 pairs from the twelve bases $\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}$ and all 50,847,532 primes up to 10^9 , the same-prime correlation $\phi(a, b)$ between the indicator variables of Artin status. Three empirical findings emerge. First, the correlation is positive for 64 of the 66 pairs (mean $\phi = 0.0352$, individual $|z|$ up to 476 — which reflects the 5×10^7 sample size rather than a large effect; the correlations explain about 0.1% of variance), despite the exclusion law: the common cause $p - 1$ dominates. The only negative pairs are (2, 10) and (3, 15), the two for which the barring character has conductor 5, the smallest occurring. Second, conditioning on the number of prime factors $\omega(p - 1)$ removes only 55% of the mean correlation, but conditioning on a finer signature of $p - 1$ — the 2-adic valuation, the divisibility of $p - 1$ by each prime up to 13, and the number of larger factors — removes 95%: the cross-base correlation is, almost entirely, a reflection of the fact that all bases see the same factorisation of $p - 1$. Third, the residual that survives fine conditioning is concentrated on the triple-completed pairs (mean absolute residual 0.018 within those pairs, or 0.008 excluding one large positive outlier, versus 0.001 for non-completed pairs), and is localised in the quadratic-character channel — identifying it as the statistical residue of the exclusion law itself, visible once the common cause is conditioned away. Finally we evaluate the joint densities predicted by the Matthews–Moree–Stevenhagen Kummer-degree heuristic for all 66 pairs: with no fitted parameters they match the measured densities to a mean relative error of 0.012%, and they reproduce the sign reversal on the entangled pairs. Both the pointwise obstruction and the Kummer-degree heuristic are governed by the same quadratic class $\text{sqf}(ab)$, which appears in the first as a congruence barring half the primes and in the second as a member of the group of quadratic classes that can reduce the degree. We do not claim these are the same phenomenon: the product class belongs to that group for every pair, whereas it lowers a degree only when its conductor divides the relevant cyclotomic level, and we exhibit pairs separating the two.

1. INTRODUCTION

Artin’s conjecture on primitive roots concerns one base at a time: for non-square $a \neq -1$, the set

$$A_a = \{p \text{ prime} : a \text{ is a primitive root mod } p\}$$

should have density given by a rational multiple of Artin’s constant, as proved under GRH by Hooley [4]. The joint distribution for *several* bases at the same prime was treated by

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Matthews [11], who computed (under GRH) the density of primes for which several prescribed integers are simultaneously primitive roots.¹ Lenstra’s Galois-theoretic framework [8] covers a single finitely generated group W with a prescribed index condition, and Lenstra explicitly lists the simultaneous problem — “the set of primes q for which both 2 and 3 are primitive roots” (his (8.17), citing Matthews) — among the sets to which his results do not directly apply. The systematic Kummer-degree treatment of the simultaneous densities is due to Moree–Stevenhagen [10], whose character-sum method reproves Matthews’ theorem; see also Moree’s survey [12]. The qualitative message of that theory is that the events $p \in A_a$ and $p \in A_b$ are *not* independent, and that the deviation from independence is governed by the interaction of the Kummer–Galois extensions attached to a and b — in modern language, by *entanglement* between them.

That framework remains actively developed. Järviniemi–Perucca–Sgobba [5] give a unified treatment of Artin-type problems for finitely generated groups in number fields; Kimmel [6] treats number-field and matrix variants (all results there conditional on GRH); Klurman–Shparlinski–Teräväinen [7] obtain results on average over the base; Sgobba [13] establishes unconditional results in cases where GRH can be circumvented; and Goldmakher–Martin–Péringuey [3] propose refined conjectures on the distribution of the multiplicative order, which bear on the structure of $p - 1$ that plays the central role in Section 5 below.

The density theory for several bases at a fixed prime is well developed: [5] treats, under GRH, the joint index map $\mathfrak{p} \mapsto (\text{Ind}_{\mathfrak{p}}(W_1), \dots, \text{Ind}_{\mathfrak{p}}(W_n))$ in full generality, of which the simultaneous primitive root problem over $\mathbb{Q}$ is the special case $H = \{(1, \dots, 1)\}$. What appears not to have been done is the empirical side: measuring these joint densities and the resulting same-prime correlations at scale, and decomposing them by conditioning on the factorisation of $p - 1$. That is our subject; we take [11, 10] as the source of the density predictions tested in Section 7. As those predictions are conditional on GRH, so is every comparison we make with them; our measurements themselves are of course unconditional.

This paper is the third in a series studying correlations of Artin status computationally at the scale of 10^9 [1, 2]. The first paper [1] measured the correlation between the Artin statuses (base 10) of *consecutive* primes and decomposed it into residue channels; the second gives a classification (companion preprint, not yet refereed) of deterministic exclusion laws for consecutive Artin primes in arbitrary bases and identified the conductor of the quadratic character χ_a as the parameter governing the statistical correlation. Here we complete the picture along the remaining axis: *fix the prime, vary the base*.

Two questions organise the paper.

(Q1) *What is deterministically forbidden?* For consecutive primes, the answer given in [2] was a set of gap classes on which the quadratic character is forced to flip. For a single prime and several bases, the answer is a congruence condition on p that bars joint Artin status on half of all primes, for every pair of bases not sharing a squarefree part:

Theorem 1 (Same-prime pair exclusion; see Theorem 3). *Let a, b be positive integers and set $d = \text{sqf}(ab)$. Then no odd prime p with $(\frac{d}{p}) = -1$ is simultaneously Artin base a and Artin base b .*

By quadratic reciprocity the excluded set is an explicit union of residue classes modulo $4d$, and whenever $d > 1$ it has density exactly $1/2$ among the odd primes. So for any two bases

¹We record one limitation of our own scholarship: [11] was verified from the publisher’s bibliographic record and from independent restatements of its main theorem, not from the original text, which the author was unable to obtain. Every other reference was checked against the published record, except the companion preprints [1, 2], which are unrefereed.

with $\text{sqf}(a) \neq \text{sqf}(b)$ — with no requirement that they sit in a special configuration, and with no third base in sight — half the primes are deterministically forbidden from being Artin for both. (If $\text{sqf}(a) = \text{sqf}(b)$ then $d = 1$, the character is principal and the statement is vacuous; see Section 2.) For example 2 and 6 are never simultaneously primitive roots for $p \equiv 5, 7 \pmod{12}$; 2 and 10 never for $p \equiv 2, 3 \pmod{5}$.

Specialising the barring character to a third base of the set gives the triple form, which was our original route to the law:

Corollary 2 (Triple exclusion; see Theorem 4). *Let a, b, c be positive integers with $\text{sqf}(c) = \text{sqf}(ab)$. Then no odd prime is simultaneously Artin base a , Artin base b , and Artin base c .*

For example, no odd prime has all of 2, 5, 10 as primitive roots; none has all of 3, 5, 15; and, since sqf is all that matters, none has all of 2, 5, 40. (The restriction to odd p is essential and not cosmetic: see the sharpness remark below.) Like the gap exclusion law of [2], both statements are elementary: they need only the multiplicativity of the quadratic character and the fact that a quadratic residue is not a primitive root. The underlying criterion is moreover classical in the density theory. Moree and Stevenhagen [10, §5, (5.2)] characterise when the local quadratic admissibility set S_2 is empty — equivalently, when the naive density vanishes — by the condition that some odd-size subset I has $\prod_{i \in I} a_i$ a square; the criterion is originally Matthews' [11]. We emphasise what that statement is and is not. Emptiness of S_2 gives an unconditional obstruction at every odd prime, and Theorem 4 is the case $n = 3$, $I = \{a, b, c\}$ of exactly that obstruction. The converse — that the set of primes is infinite, or has positive density, whenever no such relation holds — is a statement about densities and is conditional on GRH in the cited framework, as the authors are careful to distinguish. We claim no priority for the obstruction itself. What we add is an elementary pointwise formulation valid at every odd prime with no hypothesis of any kind, its extension to square bases and to 1, a machine-checked proof, and — the point of emphasis here — the observation that the third base is inessential: the obstruction is a congruence, of density 1/2, attached to the single character $\chi_{\text{sqf}(ab)}$. That character is the object the rest of the paper measures.

(Q2) *What does the joint distribution actually look like, and what explains it?* Here the data speak. We compute the exact 2×2 contingency table of $(\mathbf{1}[p \in A_a], \mathbf{1}[p \in A_b])$ over all 50,847,532 primes $p \leq 10^9$, for all 66 pairs from the base set

$$\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\},$$

chosen to contain many multiplicative triples (15 of the 66 pairs lie in a triple $\text{sqf}(ab) = \text{sqf}(c)$ within the set) alongside pairs the set does not complete. The findings, developed in Sections 4–6:

- (1) *The exclusion law is visible exactly.* For every one of the 66 pairs, the number of primes $p \leq 10^9$ that are Artin for both a and b while $\chi_a(p) \neq \chi_b(p)$ is 0, as Theorem 3 requires; on the complementary half of the primes the joint density is 0.299. The deterministic law is thus not merely proved but exhibited, on 50,847,524 primes and 3.4×10^9 pair-prime incidences.
- (2) *Positivity, nonetheless.* $\phi(a, b) > 0$ for 64 of 66 pairs, with mean 0.0352, the largest exceeding the i.i.d. benchmark scale σ_0 by a factor of 476: knowing that a is a primitive root mod p makes it *more* likely that b is, even though half of all primes are barred from having both. The common cause $p - 1$ outweighs the exclusion. The largest correlations all involve base 5 (e.g. $\phi(5, 13) = 0.0667$: $P(A_{13} \mid A_5) = 0.416$ against $P(A_{13} \mid \neg A_5) = 0.350$).
- (3) *Where the exclusion wins.* The only two negative correlations are $\phi(2, 10) = -0.0302$ and $\phi(3, 15) = -0.0303$. Both are triple-completed, but dependence alone is not the

discriminant: fifteen of the 66 pairs are triple-completed and thirteen of those are positively correlated. What singles out these two is the conductor of the barring character $\chi_{\text{sqf}(ab)}$: both have $\text{sqf}(ab) = 5$, of conductor 5 — the smallest conductor occurring among the triple-completed pairs, and the only odd prime one. A plausible reading, recorded as an observation in Section 4 rather than proved, is that a small conductor is what lets the exclusion of Theorem 3 survive the averaging over $p - 1$ instead of being diluted by it.

- (4) *The positive correlation is $p - 1$ structure, not entanglement.* Conditioning on $\omega(p - 1)$ removes 55% of the mean correlation; conditioning on a fine signature of $p - 1$ — namely $(\min(v_2(p - 1), 4), \text{the set of primes } q \leq 13 \text{ dividing } p - 1, \min(\omega_{>13}(p - 1), 3))$ — removes 95%, leaving a mean residual of 0.0019. All bases are tested against the same integer $p - 1$; a prime whose $p - 1$ is “primitive-root friendly” (few small factors, small 2-part) is friendlier for *every* base simultaneously. This common-cause structure, not any interaction between the bases, is the bulk of the correlation.
- (5) *What survives is entanglement, and it is exactly localised.* After fine conditioning, the residual correlation among *triple-completed* pairs (mean absolute residual 0.0179, or 0.0077 excluding the single large positive outlier (2, 6), with individual values reaching -0.034 for (2, 10) and -0.026 for (5, 15)) is eighteen times that of the remaining pairs (0.0010), or eightfold with (2, 6) excluded; and additionally conditioning on the quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ flattens triple-completed and remaining pairs to a common +0.005. The residual is thus localised, empirically, in the quadratic-character channel $\chi_a \chi_b = \chi_{\text{sqf}(ab)}$ — the statistical shadow of Theorem 3.
- (6) *Theory reproduces all of it.* Evaluating the Matthews–Moree–Stevenhagen Kummer-degree heuristic (3) for each pair (Section 7) predicts the measured joint densities to a mean relative error of 0.012% over the 66 pairs, with no fitted parameters, including the negative ϕ of the two entangled pairs. The entangled degree collapse in the model is precisely the character relation behind Theorem 3.

The theme of the series persists: at every level, the joint behaviour of Artin primes decomposes into a small deterministic core (here, the pair exclusion law) plus a statistical component with an identifiable arithmetic carrier (here, the shared factorisation of $p - 1$, with entanglement as a measurable second-order correction). What this paper adds to the pattern is that the deterministic core and the statistical carrier are the *same* character seen twice: $\chi_{\text{sqf}(ab)}$ bars half the primes pointwise, and reappears as the degree collapse that bends the predicted densities.

2. THE SAME-PRIME EXCLUSION LAWS

Throughout, for a positive integer a we write $d_a = \text{sqf}(a)$ for its squarefree part and $\chi_a = \left(\frac{d_a}{\cdot}\right)$ for the quadratic character attached to $\mathbb{Q}(\sqrt{a})$; thus $\chi_a(p) = +1$ exactly when a is a quadratic residue modulo p (for $p \nmid 2a$). We allow a to be a square, in which case $d_a = 1$ and $\chi_a \equiv +1$ is the principal character; this convention costs nothing and lets the statements below dispense with a non-squaredness hypothesis. We have

$$p \in A_a \implies \chi_a(p) = -1, \tag{1}$$

since a quadratic residue generates a subgroup of index at least 2 in $(\mathbb{Z}/p\mathbb{Z})^\times$.

Everything in this section follows from one line: the quadratic character is completely multiplicative, so for any positive integers a, b and any odd prime $p \nmid ab$,

$$\chi_a(p) \chi_b(p) = \left(\frac{d_a d_b}{p}\right) = \left(\frac{\text{sqf}(ab)}{p}\right) = \chi_{\text{sqf}(ab)}(p), \tag{2}$$

since $d_a d_b$ and $\text{sqf}(ab)$ differ by a square. Combining (2) with the obstruction (1) gives the following.

Theorem 3 (Pair exclusion). *Let a, b be positive integers and set $d = \text{sqf}(ab)$. Then for every odd prime p with $\left(\frac{d}{p}\right) = -1$, the prime p is not simultaneously Artin base a and Artin base b .*

Proof. If $p \mid ab$ then one of the two bases is $\equiv 0 \pmod{p}$ and so is not a primitive root; assume therefore $p \nmid ab$. By (2), $\chi_a(p)\chi_b(p) = \left(\frac{d}{p}\right) = -1$, so $\chi_a(p)$ and $\chi_b(p)$ have opposite signs and in particular one of them equals $+1$; by (1) the corresponding base is not a primitive root. $\square$

The condition $\left(\frac{d}{p}\right) = -1$ is a condition on residue classes. For $d > 1$ the character $\left(\frac{d}{\cdot}\right)$ is non-principal, of fundamental discriminant d if $d \equiv 1 \pmod{4}$ and $4d$ otherwise; either way the discriminant divides $4d$, so the excluded set is an explicit union of reduced classes modulo $4d$, and exactly half of those classes carry character -1 . By the prime number theorem in arithmetic progressions the excluded set then has density exactly $1/2$ among the odd primes. So for any two bases with $d > 1$ — no third base required, no special configuration — half of all primes are deterministically barred from being Artin for both, and Theorem 3 is a *congruence* exclusion law.

The excluded case $d = 1$ is exactly $\text{sqf}(a) = \text{sqf}(b)$, i.e. ab a perfect square, i.e. $\chi_a = \chi_b$: then $\chi_a \chi_b = \chi_a^2 \equiv +1$, nothing is ever barred, and Theorem 3 is vacuous rather than false. (For instance $a = 2, b = 8$: the two bases generate the same quadratic field, and indeed 2 and 8 are simultaneously primitive roots for $p = 3, 5, 11, 29, 53, \dots$) Such pairs are precisely the ones our base set avoids: all twelve bases are squarefree and distinct, so $d > 1$ for all 66 pairs. Writing the classes out in the four cases we return to below:

$$\begin{aligned} (2, 6), d = 3: & \quad p \equiv 5, 7 \pmod{12}; & (3, 6) \text{ and } (5, 10), d = 2: & \quad p \equiv 3, 5 \pmod{8}; \\ (2, 10), d = 5: & \quad p \equiv 2, 3 \pmod{5}. \end{aligned}$$

We verified each over all primes below 6×10^4 with no exceptions, and Section 4 verifies the theorem itself against all 50,847,524 primes below 10^9 for all 66 pairs of our base set simultaneously. Theorems 3 and 4 are also machine-checked in Lean 4 against Mathlib, including the step (1) and the (5, 10) instance above, whose barring condition is discharged from $p \pmod{8}$ — so that instance is verified with no character hypothesis assumed at all. Every checked statement depends on no axiom beyond `propext`, `Classical.choice` and `Quot.sound`. We state the limit of this plainly: the general translation of $\left(\frac{d}{p}\right) = -1$ into residue classes modulo $4d$ is *not* formalised generically, but carried out by hand per instance. See `LEAN_NOTE.md` in the accompanying package.

The triple form of the law, which was our original route to it, is the special case in which the barring character is supplied by a third base of the set.

Theorem 4 (Triple exclusion). *Let a, b, c be positive integers with $\text{sqf}(c) = \text{sqf}(ab)$ and let p be an odd prime. Then p is not simultaneously Artin base a , Artin base b , and Artin base c .*

Proof. If p is Artin base c then $\chi_c(p) = -1$ by (1), and $\chi_c = \chi_d$ with $d = \text{sqf}(ab) = \text{sqf}(c)$; now apply Theorem 3. $\square$

Corollary 5. *Within any multiplicative triple — for instance $(2, 5, 10)$, $(3, 5, 15)$, $(3, 7, 21)$, $(2, 3, 6)$ — an odd prime can be Artin for at most two of the three bases. The restriction to odd p is necessary and not cosmetic: every odd integer is vacuously a primitive root modulo 2, so all three members of $(3, 5, 15)$ are Artin base 2.*

The gain in passing from Theorem 4 to Theorem 3 is not cosmetic, though it is a gain in usable content rather than in logical strength: both are pointwise statements at every odd

prime. The difference is in what one must know to apply them. To invoke Theorem 4 one must first identify a third base c with $\text{sqf}(c) = \text{sqf}(ab)$ and establish that c is a primitive root at the prime in question — itself an instance of the Artin problem, conjectural in general and (under GRH) of positive but base-dependent density. Theorem 3 needs neither: the barring condition is a congruence, decidable from p alone by reciprocity, and for $d > 1$ it holds on a set of density exactly $1/2$. The triple law is the shadow that the pair law casts when one insists on reading the barring character off a primitive root instead of off a residue class.

Remark 6 (Sharpness of the hypotheses). In both theorems only “ p odd” is needed. Non-squareness of the bases is not required: if a is a square then $\chi_a(p) = +1$ and a is not a primitive root, so the conclusion is immediate. The same remark disposes of $c = 1$ in Theorem 4, which arises whenever $\text{sqf}(a) = \text{sqf}(b)$. The hypothesis $p \neq 2$ must however be kept, and is not removable: $(\mathbb{Z}/2\mathbb{Z})^\times$ is trivial, so every odd integer is vacuously a primitive root mod 2 and e.g. $(a, b, c) = (3, 5, 15)$ violates the conclusion there. The accompanying Lean formalisation (Section 3) confirms independently that the $a \not\equiv 0$ and $b \not\equiv 0 \pmod{p}$ assumptions are never used.

Remark 7. The exclusion is sharp: primes Artin for exactly two of $(2, 5, 10)$ exist in abundance. The smallest is $p = 3$, where 2 and 5 are both primitive roots (each of order $2 = p - 1$) while $10 \equiv 1$ is not; the smallest for which 10 itself is Artin is $p = 7$, where 5 and 10 are both primitive roots (each of order 6) while 2 has order 3; likewise $p = 23$, where 5 and 10 are primitive roots and 2 is a quadratic residue of order 11. More systematically, in our data the pair $(2, 10)$ has $P(A_{10} \mid A_2) = 0.355$: two-of-three is common, three-of-three is impossible.

Remark 8. Theorem 3 is the same-prime analogue of the gap exclusion law for consecutive primes proved in [2]: both are instances of a parity obstruction in the group generated by the quadratic characters involved, and in both the excluded set is a union of residue classes. There the two characters were χ_a evaluated at p and at $p + g$, and the obstruction was arranged by the shift g ; here the characters belong to different bases at the same prime, and the obstruction is arranged by multiplicative dependence. The Galois-theoretic density computations of Matthews [11] and Moree–Stevenhagen [10] contain this relation as the vanishing criterion recalled in Section 1 (the degree of the compositum of the relevant Kummer extensions drops); the elementary pointwise form seems worth recording.

3. DATA AND COMPUTATION

By a segmented sieve we enumerated all primes $p \leq 10^9$ (50,847,534 primes). The programs discard only the smallest primes, for which the 128-bit Euler-criterion arithmetic and the signature encoding are degenerate: $p < 5$ for the correlation and decomposition runs, leaving 50,847,532 primes, and $p < 31$ for the signature-and-character run of Section 6, leaving 50,847,524. A prime dividing a base is *not* discarded; it is simply recorded as non-Artin for that base, which is correct, since 0 is not a primitive root. The two counts therefore differ by exactly the eight primes in $[5, 31)$. For each prime we factored $p - 1$ once by trial division and computed the Artin indicator for each of the twelve bases by the standard criterion ($a^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime $\ell \mid p - 1$, in 128-bit arithmetic). One streaming pass accumulates, for each of the 66 unordered base pairs, the 2×2 contingency table of joint Artin status — globally, and stratified by (i) $\omega(p - 1)$ and (ii) the fine signature

$$\text{sig}(p) = \left(\min(v_2(p - 1), 4), \{q \leq 13 : q \mid p - 1\}, \min(\omega_{>13}(p - 1), 3) \right),$$

a partition of the primes into at most $4 \cdot 32 \cdot 4 = 512$ cells (for odd p we always have $v_2(p - 1) \geq 1$, so the truncated valuation takes four values, not five). The signature records

exactly the information about $p - 1$ that the Artin criterion consults for small bases: the 2-part and the small odd prime divisors. A separate run, also to 10^9 (50,847,524 primes), stratified each pair additionally by the quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ of the two bases on top of the signature. The whole computation runs in under two hours on a single core of a commodity machine (2-core Intel Xeon E5-2673 v4, 8 GB RAM); all counts are exact.

As validation, the marginal Artin densities agreed with the Hooley densities to within 6×10^{-5} absolute (0.014% relative) for every base — e.g. 0.373957 for base 2, against Artin’s constant 0.373956, and 0.393642 for base 5, whose density carries the classical correction factor $C \cdot \frac{20}{19}$ for $d \equiv 1 \pmod{4}$ (see [4, 12]).

4. THE CROSS-BASE CORRELATION

For each pair write $\phi(a, b)$ for the Pearson correlation of the indicator variables (the ϕ coefficient of the 2×2 table). This is an exhaustive census of a fixed interval for a deliberately chosen finite set of bases, so the coefficients below are exact and carry no sampling error. We therefore report effect sizes and their variation with x , and avoid the vocabulary of statistical inference. Where a scale factor is quoted we write $\sigma_0 = N^{-1/2} \approx 0.00014$ at $N \approx 5.08 \times 10^7$: this is the standard deviation of ϕ under an i.i.d. independence benchmark that does *not* describe these data — the primes are enumerated, not sampled — and ratios ϕ/σ_0 should be read as model-relative diagnostics indicating that the effects are far larger than that benchmark’s scale, not as tests of any hypothesis. No claim below depends on such a ratio.

One convention should be stated, since it affects the conditional averages in Sections 5 and 6. Within-cell ϕ is undefined when a cell is constant in one of the two indicators (for example cells with $5 \mid p - 1$, in which no prime is Artin base 5). Such cells carry zero conditional covariance and are omitted from the pooled average, which therefore draws on 73% of primes on average for the signature stratification, and 23% for the finer signature-plus-character stratification, where three of the four character quadrants are identically non-Artin. Explicitly, for a base pair (a, b) we report

$$\phi \mid \text{sig} = \frac{\sum_{s \in E_{ab}} n_s \phi_{ab,s}}{\sum_{s \in E_{ab}} n_s},$$

where s runs over signature cells, n_s is the number of primes in cell s , and E_{ab} is the set of cells with $n_s \geq 200$ in which *both* indicators vary. This is a prime-count-weighted descriptive mean of within-cell correlations, not a partial correlation coefficient and not an additive component of ϕ ; outer means over the 66 base pairs are unweighted. Pooling instead on the covariance scale over all cells, which requires no such exclusion, gives 95.3% removal, in agreement with Table 3.

Three observations, in increasing order of depth.

Observation 9 (Positivity). $\phi(a, b) > 0$ for 64 of the 66 pairs. Artin statuses of different bases at the same prime *attract*.

The sign is the opposite of the consecutive-prime correlation measured in the first two papers of the series [1, 2] (where $\delta < 0$ uniformly), and the mechanism is transparent: there the two primes had *different* (and LOS-anticorrelated) values of $p - 1$; here every base is tested against the *same* $p - 1$. If $p - 1$ has few small prime factors and a small 2-part, the primitive-root condition is easier for every base at once. Formally, the Artin indicators share the common cause $\text{sig}(p)$.

Observation 10 (The negative pairs are the entangled ones). The only pairs with $\phi < 0$ are (2, 10) and (3, 15): in each, the product of the two bases is, up to a square, the third base of a multiplicative triple contained in the set $(2 \cdot 10 = 4 \cdot 5, 3 \cdot 15 = 9 \cdot 5)$. For (2, 10): $P(A_{10} \mid A_2) = 0.3551$ versus $P(A_{10} \mid \neg A_2) = 0.3853$.

TABLE 1. Same-prime cross-base Artin correlation at $x = 10^9$: summary and extremes. “Triple-completed” means the base set contains a third base c with $\text{sqf}(c) = \text{sqf}(ab)$. This is a property of the chosen base set, not of the pair: the two bases of such a pair are multiplicatively independent in the usual sense, and adding or removing c changes the label but not the pair’s density, character or Kummer field.

	pairs	mean ϕ
all pairs	66	+0.0352
non-completed	51	+0.0363
triple-completed	15	+0.0314
largest: (5, 13)		+0.0667
smallest: (3, 15)		−0.0303
second smallest: (2, 10)		−0.0302

TABLE 2. The ten largest $|\phi|$ at $x = 10^9$, with the values after conditioning on $\omega(p-1)$ and on the fine signature $\text{sig}(p)$ (pair-weighted mean of within-cell ϕ ; cells with fewer than 200 primes excluded). “Dep.” marks triple-completed pairs.

a	b	ϕ	$\phi \mid \omega$	$\phi \mid \text{sig}$	Dep.
5	13	+0.0667	+0.0200	+0.0003	
5	17	+0.0649	+0.0218	+0.0025	
5	29	+0.0631	+0.0246	+0.0007	
5	10	+0.0624	+0.0285	−0.0125	✓
5	6	+0.0623	+0.0292	+0.0005	
5	11	+0.0623	+0.0300	+0.0004	
5	15	+0.0622	+0.0297	−0.0255	✓
3	5	+0.0622	+0.0284	+0.0015	✓
5	7	+0.0622	+0.0297	+0.0002	
2	5	+0.0621	+0.0336	+0.0004	✓

The sign reversal is Theorem 3 operating statistically. For every pair, triple-completed or not, the primes split into two halves according to the sign of $\chi_d(p)$, $d = \text{sqf}(ab)$: on the half with $\chi_d(p) = -1$ the values $\chi_a(p)$ and $\chi_b(p)$ are forced *opposite*, so joint Artin status is impossible there, while on the half with $\chi_d(p) = +1$ they are forced *equal* and joint Artin status is unobstructed. This is visible directly in the data: over all 66 pairs and all 50,847,524 primes, the number of primes Artin for both bases with $\chi_a(p) \neq \chi_b(p)$ is exactly zero, while the conditional joint density on the complementary half is 0.299.

What then distinguishes the two negatively correlated pairs is not the existence of the exclusion — every one of the 66 pairs has it, on exactly half the primes — but the identity of the barring character χ_d , and in particular how its conductor interacts with the common cause $p-1$. Among the fifteen triple-completed pairs, $d = \text{sqf}(ab)$ takes the values $\{2, 3, 5, 6, 7, 10, 15, 21\}$, and the two pairs with $d = 5$ — (2, 10) and (3, 15), of conductor 5, the smallest conductor occurring and the only odd prime one — are exactly the two with $\phi < 0$. That is the sharp empirical discriminant: not dependence, which thirteen positively correlated pairs also enjoy, but

TABLE 3. Mean cross-base correlation after conditioning, all 66 pairs at $x = 10^9$. “Removed” is $1 - |\phi_{\text{cond}}|/|\phi|$, the means taken signed over the 66 pairs; on the mean-absolute scale the corresponding figures are 51% and 87%, and on the covariance scale 57% and 95%.

conditioning	mean ϕ	removed
none	0.0352	—
$\omega(p-1)$	0.0158	55%
$\text{sig}(p)$: $(v_2, \{q \leq 13\}, \omega_{>13})$	0.0019	95%

$\text{cond}(\text{sqf}(ab)) = 5$. A plausible reading is that a small conductor makes the barring condition $\chi_d(p) = -1$ resolve on a coarse residue class that is comparatively well correlated with the divisibility structure of $p-1$ driving the common cause, so the exclusion survives the averaging instead of being diluted by it; we do not have a proof of this and record it as an observation. Section 6 tests the reading by conditioning the common cause away, after which the exclusion becomes visible on six triple-completed pairs rather than two.

5. DECOMPOSITION: THE CORRELATION IS THE STRUCTURE OF $p-1$

Conditioning on $\omega(p-1)$ alone — the crude size of the factorisation — removes barely half of the correlation. This is worth emphasising because ω is the statistic through which the consecutive-prime papers routed the mechanism, and it is genuinely insufficient here: two primes with the same $\omega(p-1)$ can present very different Artin conditions (it matters *which* primes divide $p-1$, and with what 2-adic valuation, not how many). The fine signature — 2-adic valuation capped at 4, the exact set of primes ≤ 13 dividing $p-1$, and the count of larger prime factors capped at 3 — is the information the Artin criterion actually consults for small bases, and conditioning on it removes 95% of the mean correlation (Table 3). That the fine structure of $p-1$, rather than its coarse size, is what governs Artin behaviour is in the same spirit as the refined order-distribution conjectures of [3], which pass from the Artin condition $\text{ord}_p(a) = p-1$ to the distribution of $\omega((p-1)/\text{ord}_p(a))$ — finer information about the index than its mere vanishing, just as our signature is finer information about $p-1$ than $\omega(p-1)$.

We regard this as the main statistical finding: *cross-base Artin correlation at a single prime is, to first order, not an interaction between the bases at all*. It is a common-cause effect of the shared integer $p-1$. Any model that treats A_a and A_b as conditionally independent given the small-prime structure of $p-1$ reproduces 95% of the observed association.

6. THE RESIDUAL AND THE EXCLUSION LAW

The 5% that survives is not uniformly distributed: it is concentrated on the triple-completed pairs. This section argues that it is the statistical trace of Theorem 3 — the deterministic exclusion that the unconditional correlation of Section 4 had largely averaged away.

Two facts pin down the mechanism, both measured over the full 10^9 range.

First, after fine conditioning the triple-completed pairs retain substantial residuals, and every pair with a substantial *negative* residual is triple-completed. The six pairs below are the only ones with $\phi \mid \text{sig} < -0.004$, and all six are triple-completed. We state this as the exact combinatorial fact it is — six of the six most negative residuals fall in the triple-completed class — rather than attaching a p -value to it: the 66 pairs share endpoints and the base set was chosen to contain multiplicative triples, so the pair labels are neither independent nor exchangeable

TABLE 4. Residual correlation after conditioning on $\text{sig}(p)$ and after additionally conditioning on the pair of quadratic-residue statuses $(\chi_a(p), \chi_b(p))$, split by multiplicative dependence. All values at $x = 10^9$ over 50,847,524 primes.

group	mean ϕ	mean $\phi \mid \text{sig}$	mean $\phi \mid \text{sig, QR}$
triple-completed (15 pairs)	+0.0314	+0.0049	+0.0049
mean <i>absolute</i> residual		0.0179	
non-completed (51 pairs)	+0.0363	+0.0010	+0.0052
mean <i>absolute</i> residual		0.0010	
all 66 pairs	+0.0352	+0.0019	+0.0052

and no randomisation model licences such a test.

$$\begin{aligned}
 (2, 10): -0.0343, & \quad (5, 15): -0.0255, & \quad (6, 10): -0.0144, \\
 (5, 10): -0.0125, & \quad (7, 21): -0.0060, & \quad (10, 15): -0.0049.
 \end{aligned}$$

At $x = 10^9$ these are also the only six negative residuals of any size, but that sharper statement is not stable with x : at $x = 3 \times 10^7$ a further twelve pairs have residuals marginally below zero, all within $3\sigma_0$ of it in the benchmark scale of Section 4.

The residual is not sign-definite, and we record the exceptions explicitly. Nine of the fifteen triple-completed pairs have *positive* residuals: eight of them small, and one, $(2, 6)$, a large outlier at $+0.1598$ — the largest residual of any pair in the collection. Its mechanism is transparent and is the same entanglement phenomenon seen from the other side: $\text{sqf}(6) = 2 \cdot 3$. The relevant family is narrower than $3 \nmid p - 1$ alone, however. If $3 \nmid p - 1$ and $v_2(p - 1) \geq 2$ then $p \equiv 5 \pmod{12}$, so $(3/p) = -1$ and Theorem 3 *forbids* 2 and 6 together. Agreement can occur only on the complementary family $v_2(p - 1) = 1$, that is $p \equiv 11 \pmod{12}$, where $(3/p) = +1$ and the two quadratic characters coincide; that the remaining odd-prime tests then also very nearly coincide is what we measure, not something the factorisation $\text{sqf}(6) = 2 \cdot 3$ implies on its own, and the signature records whether $3 \mid p - 1$ without recording the resulting near-determinism inside those cells. (The effect is sharpest in the cells where $p - 1$ has no small odd prime factor at all: for $v_2(p - 1) = 1$, no prime $q \leq 13$ dividing $p - 1$ and exactly one larger factor, the 2×2 table for $(2, 6)$ is exactly diagonal over 1,775,686 primes, giving $\phi = +1$. Across the 44 cells with $v_2(p - 1) = 1$ and $3 \nmid p - 1$ the within-cell ϕ ranges from $+0.39$ to $+0.98$, so the near-determinism is a feature of that family of cells rather than of a single one.) The triple-completed-group mean absolute residual of 0.0179 is substantially driven by this one pair: excluding it, the figure is 0.0077, still eightfold the value for the remaining pairs of 0.0010. Conditioning additionally on $(\chi_a(p), \chi_b(p))$ reduces $(2, 6)$ to $+0.0055$, in line with every other pair — so the outlier is explained by the entanglement account rather than in tension with it. Note that $(2, 10)$ and $(5, 15)$ are more negative *after* conditioning than before, and that $(5, 10)$, $(5, 15)$, $(6, 10)$, $(10, 15)$ and $(7, 21)$ change sign under conditioning: the common-cause channel had been masking an underlying repulsion.

Second, additionally conditioning on the pair of quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ removes the negative residuals. We put it this way deliberately: the signed group means do not both fall to zero. The triple-completed mean stays at $+0.0049$ while the mean for the remaining pairs *rises* from $+0.0010$ to $+0.0052$, and the two averages are taken over different eligible populations (about 73% of primes for signature conditioning against 23% here), so they are not components of a single decomposition. Every one of the six negative values above becomes positive and small ($+0.0031$, $+0.0066$, $+0.0045$, $+0.0065$, $+0.0040$, $+0.0048$ respectively), and

the triple-completed and remaining groups become indistinguishable at $+0.0049$ and $+0.0052$ (Table 4). That common small positive floor of about $+0.005$ is present for all 66 pairs and is plausibly attributable to residual odd-prime conditions shared through $p - 1$ beyond what our signature records. We note that a *quartic* channel is not available as an explanation here: a fourth power is a square, and in the only quadrant contributing defined correlations both bases are quadratic nonresidues, so no further 2-power residue condition can bind.

Conditioning on the quadratic characters is precisely what removes character-level entanglement, and that it flattens the triple-completed pairs is consistent with the residual as the statistical trace of the relation $\chi_a \chi_b = \chi_{\text{sqf}(ab)}$ — the same object that makes Theorem 3 true.

One part of this can be made exact, though it is a statement about support and not about the residual's size. Theorem 3 asserts that the joint Artin event is empty on the stratum $\chi_a(p) \neq \chi_b(p)$, and this is what the contingency tables show: summing the joint-Artin cell over all signature classes and all 66 pairs, the count of primes $p \leq 10^9$ that are Artin for both a and b with $\chi_a(p) \neq \chi_b(p)$ is

$$0 \quad (\text{out of } 50,847,524 \text{ primes, for every one of the 66 pairs}),$$

while on the complementary stratum the joint density is 0.299. The support restriction is therefore a theorem rather than a conjecture. It does not by itself determine the sign or the magnitude of any residual: a support constraint of this kind is compatible with zero covariance, as the trivial example of two independent fair signs with Artin indicators given by their negative-sign indicators shows. What the residual measures is a quadratic-information component; that this component is the pointwise law's statistical trace is an interpretation the data support, not a consequence of the theorem.

One caveat on the strength of the flattening step, in the other direction. Within a (χ_a, χ_b) stratum the exclusion cannot operate at all: by criterion (1) an Artin prime for a has $\chi_a(p) = -1$, so all the *joint* Artin mass sits in the single quadrant $(-1, -1)$. In each of the other three quadrants at least one of the two indicators is identically zero, so the within-cell correlation is undefined and the conditional covariance vanishes; this is not the same as saying neither indicator varies there. Conditioning on (χ_a, χ_b) therefore *restricts attention to the stratum on which the constraint of Theorem 3 is vacuous*, and the vanishing of the residual there is in part a structural consequence of that restriction rather than an independent confirmation. What the step establishes is the localisation: the residual lives in the character channel and not elsewhere in the signature. We state it in that weaker form.

The picture that emerges is cleanly layered:

$$\underbrace{\text{common cause } p-1}_{\approx 95\%} + \underbrace{\text{quadratic entanglement } \chi_a \chi_b = \chi_{\text{sqf}(ab)}}_{\text{residual, localised on triple-completed pairs}} + \underbrace{\text{higher-order sharing}}_{\text{small positive floor}}.$$

7. COMPARISON WITH THE PREDICTED JOINT DENSITIES

The measurements of Sections 4–6 identify the mechanisms empirically. We now check them against theory quantitatively, by computing the joint densities predicted by the Matthews–Moree–Stevenhagen framework and comparing with the observed counts pair by pair.

Write $d_a = \text{sqf}(a)$. The heuristic, in the form originating with [11] and systematised by the character-sum method of [10] (with the general framework in [5]), is an inclusion–exclusion over the degrees of the Kummer extensions that detect the conditions “ a is an ℓ -th power residue”:

$$\mathbf{P}(p \in A_a \cap A_b) = \sum_{m,n \text{ squarefree}} \frac{\mu(m) \mu(n)}{[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}]}, \quad L = \text{lcm}(m, n), \quad (3)$$

TABLE 5. Predicted versus measured joint Artin densities, selected pairs at $x = 10^9$. “Dep.” marks triple-completed pairs. The final two rows are the two negatively correlated pairs.

a	b	measured j	model j	rel. err.	measured ϕ	model ϕ	Dep.
5	13	0.163939	0.163943	+0.00%	+0.0667	+0.0667	
5	10	0.161962	0.161922	−0.02%	+0.0624	+0.0623	✓
5	15	0.161934	0.161922	−0.01%	+0.0622	+0.0623	✓
3	7	0.149965	0.149971	+0.00%	+0.0432	+0.0433	✓
13	17	0.150249	0.150261	+0.01%	+0.0383	+0.0383	
2	3	0.147361	0.147349	−0.01%	+0.0321	+0.0321	✓
2	6	0.147334	0.147349	+0.01%	+0.0320	+0.0321	✓
7	21	0.144754	0.144728	−0.02%	+0.0238	+0.0238	✓
15	21	0.144725	0.144728	+0.00%	+0.0237	+0.0238	
3	15	0.132769	0.132776	+0.01%	−0.0303	−0.0302	✓
2	10	0.132776	0.132776	+0.00%	−0.0302	−0.0302	✓

and the entanglement enters through the degree in the denominator. Generically

$$[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}] = \varphi(L) m n,$$

but the degree drops by the order of the subgroup of quadratic classes realised inside $\mathbb{Q}(\zeta_L)$, taken from the group

$$V(m, n) = \langle d_a \text{ (if } 2 \mid m), d_b \text{ (if } 2 \mid n) \rangle \subseteq \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$$

whose quadratic field lies inside $\mathbb{Q}(\zeta_L)$, i.e. whose conductor divides L . The point of contact with Theorem 3 is that when both $2 \mid m$ and $2 \mid n$ the group V contains not only d_a and d_b but their product $d_a d_b \sim \text{sqf}(ab)$, so the barring character $\chi_{\text{sqf}(ab)}$ of Theorem 3 is present in the model as a potential source of degree collapse. Two things must be said precisely here, because a natural stronger reading is false. First, the product class lies in $V(m, n)$ for *every* pair of bases whenever both indices are even; its membership is automatic and has nothing to do with whether a third base of our set completes a multiplicative triple. Second, membership in V is not collapse: a class contributes only when its conductor divides L . These two conditions do not coincide, in either direction. For (7, 11) — a pair outside every triple of our set — the product class 77 has conductor 77, which divides $L = 154$, while the individual conductors 28 and 44 do not, so the product class does contribute a collapse there. Conversely for (2, 6) — a pair inside the triple (2, 3, 6) — the nontrivial classes of V have conductors 8, 24 and 12, all divisible by 4, and no squarefree L is, so no term of (3) receives this correction at all. The triple-completion classification used in our tables is therefore a property of the chosen base set, not a field-theoretic invariant of the pair, and it is not equivalent to the occurrence of an extra degree collapse. We evaluate (3) by summing the ramified part exactly over all squarefree m, n supported on $S = \{2\} \cup \{q : q \mid d_a d_b\}$ and folding the rest into the Euler product $\prod_{q \notin S} (1 - \frac{2q-1}{q^2(q-1)})$, truncated at 3×10^6 .

As a check on the implementation, the same code specialised to a single base reproduces the classical Hooley densities: 0.373956 for bases with $d \not\equiv 1 \pmod{4}$ (Artin’s constant), 0.393638 for $d = 5$, 0.376368 for $d = 13$, against measured 0.373957, 0.393642, 0.376362.

Observation 11 (The model reproduces the measurements). Over all 66 pairs at $x = 10^9$, the mean absolute relative error between the predicted and measured joint densities is 0.012%, with

maximum 0.040% (attained at (21, 29)). In particular the model predicts the sign reversal: for the entangled pairs (2, 10) and (3, 15) it gives $j = 0.132776$ and $\phi = -0.0302$ for both, against measured $j = 0.132776, 0.132769$ and $\phi = -0.0302, -0.0303$ respectively.

This is the quantitative counterpart of Sections 4 and 6. It is a comparison of unconditional joint densities, so it does not by itself validate any conditional mechanism, but it is a demanding test and the framework passes it. The 66-pair pattern — positive correlation almost everywhere, negative precisely on the two entangled pairs, with magnitudes ordered as observed — is not merely *consistent with* the Matthews–Moree–Stevenhagen framework; it is reproduced by it to within a few parts in 10^4 , with no fitted parameters. Correspondingly, the same quadratic class $\text{sqf}(ab)$ governs the deterministic law of Theorem 3 and enters the heuristic through the group $V(m, n)$. We stop short of calling these one object: the class lies in $V(m, n)$ for every pair once both indices are even, while it lowers a degree only when its conductor divides L , and the two conditions come apart in both directions as noted in Section 7.

Remark 12. The residual discrepancies are uniformly small (all 66 pairs below 0.040%) and we have not pursued them; ordinary Hooley-type errors of this size are expected at $x = 10^9$, since the convergence in Hooley’s theorem is only logarithmic, and our Euler products are truncated at 3×10^6 . We record one methodological warning, since it cost us a draft. Computing $\varepsilon(m, n)$ as 2 raised to the number of classes of $V(m, n)$ with conductor dividing L — rather than as the order of the subgroup of $V(m, n)$ realised in $\mathbb{Q}(\zeta_L)$ — inflates ε to 8 on exactly those pairs with $d_a \equiv d_b \equiv 1 \pmod{4}$, where all three of $d_a, d_b, \text{sqf}(ab)$ have odd conductor. Since $V(m, n)$ has two generators, $|V(m, n)| \leq 4$ and $\varepsilon = 8$ is impossible. The error is invisible in the single-base specialisation, which is why the check above does not catch it; it inflates the mean relative error over the 66 pairs from 0.012% to 0.037% and concentrates the damage on the ten pairs drawn from $\{5, 13, 17, 21, 29\}$.

8. DISCUSSION

Relation to the first two papers of the series [1, 2]. The trilogy now covers both axes of the joint distribution of Artin status. Along the prime axis (consecutive p_n, p_{n+1} , one base), correlation is negative, driven by the Lemke Oliver–Soundararajan bias [9] propagating through the two distinct integers $p_n - 1$ and $p_{n+1} - 1$, with deterministic exclusion classes in the gap. Along the base axis (one prime, two bases), correlation is positive for 64 of the 66 pairs measured, driven by the single shared integer $p - 1$, with deterministic exclusion on an explicit half of all primes for every pair of bases with distinct squarefree parts. In both cases the deterministic law is a parity statement about quadratic characters attached to a congruence condition — there on the gap $p_{n+1} - p_n$, here on p itself — and in both cases it is statistically dominated by a factorisation-of- $p - 1$ mechanism: diluted to invisibility in the global average there, absorbed almost entirely by conditioning here. The symmetry is closer than we expected when the series began: what looked like a statement about special configurations of bases (multiplicative triples) turned out, once the third base was seen to be inessential, to be a statement of the same shape as the gap law — a character condition on a residue class of density $1/2$.

Future work. (1) A rigorous version of Section 7: the comparison there is between measurement and a heuristic evaluated numerically, and the error terms — both the Hooley-type error at $x = 10^9$ and the truncation of our Euler tail — are not controlled. Making the agreement into a theorem (conditionally on GRH, in the manner of [11]) would require uniform error bounds we do not attempt. (2) The conductor dependence made precise. Theorem 3 bars half the primes for every pair of our base set, yet only the two pairs with $\text{cond}(\text{sqf}(ab)) = 5$ show a negative unconditional correlation; Section 4 offers a heuristic reason but no proof. Quantifying the

surviving exclusion as a function of $\text{cond}(\text{sqf}(ab))$ — equivalently, computing the covariance between the barring condition $\chi_{\text{sqf}(ab)}(p) = -1$ and the signature of $p - 1$ — looks tractable within the Kummer-degree framework and would turn the empirical discriminant into a formula. A natural test is to enlarge the base set so that several small odd prime conductors occur among the triple-completed pairs, which our twelve bases do not provide. (3) The mixed axis: ϕ between $A_a(p_n)$ and $A_b(p_{n+1})$ for $a \neq b$ — the data pipeline extends immediately, and the prediction from the present findings is a small negative correlation governed by the two conductors jointly. (4) Higher-order entanglement: triples of bases conditioned on quadratic data should expose the cubic layer ($\ell = 3$ Kummer conditions), with $\mathbb{Q}(\zeta_3)$ playing the role $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{2})$ played here. There is no direct analogue of Theorem 3 at $\ell = 3$. The quadratic argument works because -1 is the only nonidentity value and the product of two such values is forced to be $+1$; for cubic characters the nonidentity values ζ_3, ζ_3^2 have products realising all three values, so requiring both bases to be cubic nonresidues imposes no condition on their product. Even a relation $\text{sqf}(c) = \text{sqf}(ab)$ is compatible with the assignment $(\zeta_3, \zeta_3, \zeta_3^2)$. Whether some higher-order Frobenius relation produces an obstruction, and over which population of primes (cubing is a bijection when $p \equiv 2 \pmod{3}$), is a question we leave open rather than a stated analogue.

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DECLARATION OF INTEREST

The author reports there are no competing interests to declare.

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DATA AVAILABILITY

All code and results are publicly available at [repositoryURLwithheldforreview](#), including the single-pass multi-base programs (`crossbase.c`, `crossbase_fine.c`, `crossbase_om.c`, `crossbase_qr.c`), the analysis scripts, and the raw contingency-table output at $x = 10^9$ from which every value quoted here is derived, namely

`crossbase_qr_1e9.json`, `crossbase_fine_1e9.json`, `crossbase_om_1e9.json`

(3.2 MB in total). Running `analyze_qr.py` on the first of these reproduces Table 4 exactly. The full computation reproduces in under two hours on a single core; the $x = 10^9$ signature-and-character run alone takes about 16 minutes on 1 core of a modern desktop. The Lean

sources for the machine-checked statements are included in the same repository, with build instructions in `BUILD.md`.

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